is used to let the fuel to go back to the fuel tank. Figure 6.9 shows the Fuel Injector.

Working Principle of Injectors

By the rotation of engine crankshaft, the camshaft rotates by using the timing gear connected with it. Hence the camshaft connected to the fuel injection pump also rotates and high pressure is developed in the fuel injection pump. The pressurized fuel the fuel tank is supplied to the fuel injector through the high-pressure fuel line. The diesel fuel entering the injector reaches the nozzle and starts to spray with the help of spring force acting inside the nozzle. Diesel fuel enters into the nozzle at very high pressure opens the nozzle valve by lifting against spring force acting inside the spindle. Hence the nozzle valve opens and high-pressure diesel fuel in the nozzle is injected into the chamber generally at the pressure of 200 bars in the form of very fine droplets. The droplets of the diesel fuel then vaporize and mix with the air inside the combustion chamber to attain proper mixture for burning. When the injection process is over the pressure getting reduced in the spring inside the nozzle, hence the spindle moves the nozzle valve downward. Hence the nozzle valve gets closed. In this way the diesel fuel injector works.

6.10 NOZZLE

Nozzle is the important component used for injecting the diesel at very high pressure at correct timing for the maximum power output of the engine. It is a part of injector which is placed at the cylinder head and has contact with combustion chamber in order to inject the fuel. It helps in injecting the fuel as per the requirements

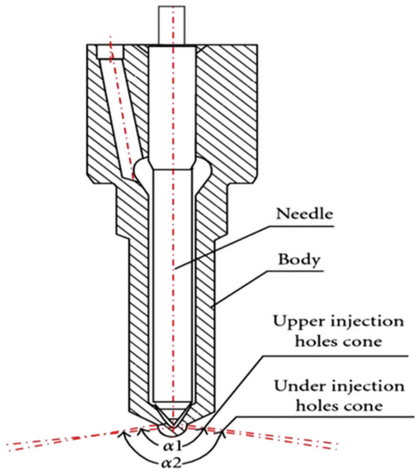


Figure 6.10 Nozzle

of the engine at different pressures and at constant temperature with accurate metering. Nozzle is designed to fit in to the nozzle valve and it can be assembled only by the authorized company. A hole for supplying fuel is available at the tip of nozzle valve. The diesel entering into the injector at high pressure is used for opening the valve and the returning action is taken by springs provided in the valve. Figure 6.10 shows the Nozzle.

There are different types of nozzles,

1. Single Hole Nozzle
2. Multi Hole Nozzle
3. Long Stem Nozzle

6.10.1 Single Hole Nozzle

It has a single hole at the tip for injecting a fuel. The nozzle has a cone shaped spindle which is used for opening and closing the nozzle hole. Figure 6.10.1 shows the Single Whole nozzle.

6.10.2 Multi Hole Nozzle

It has a multi holes for injecting the fuel depending upon the engine

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